



ПРЕЗИДЕНТСКАЯ
АКАДЕМИЯ

2026

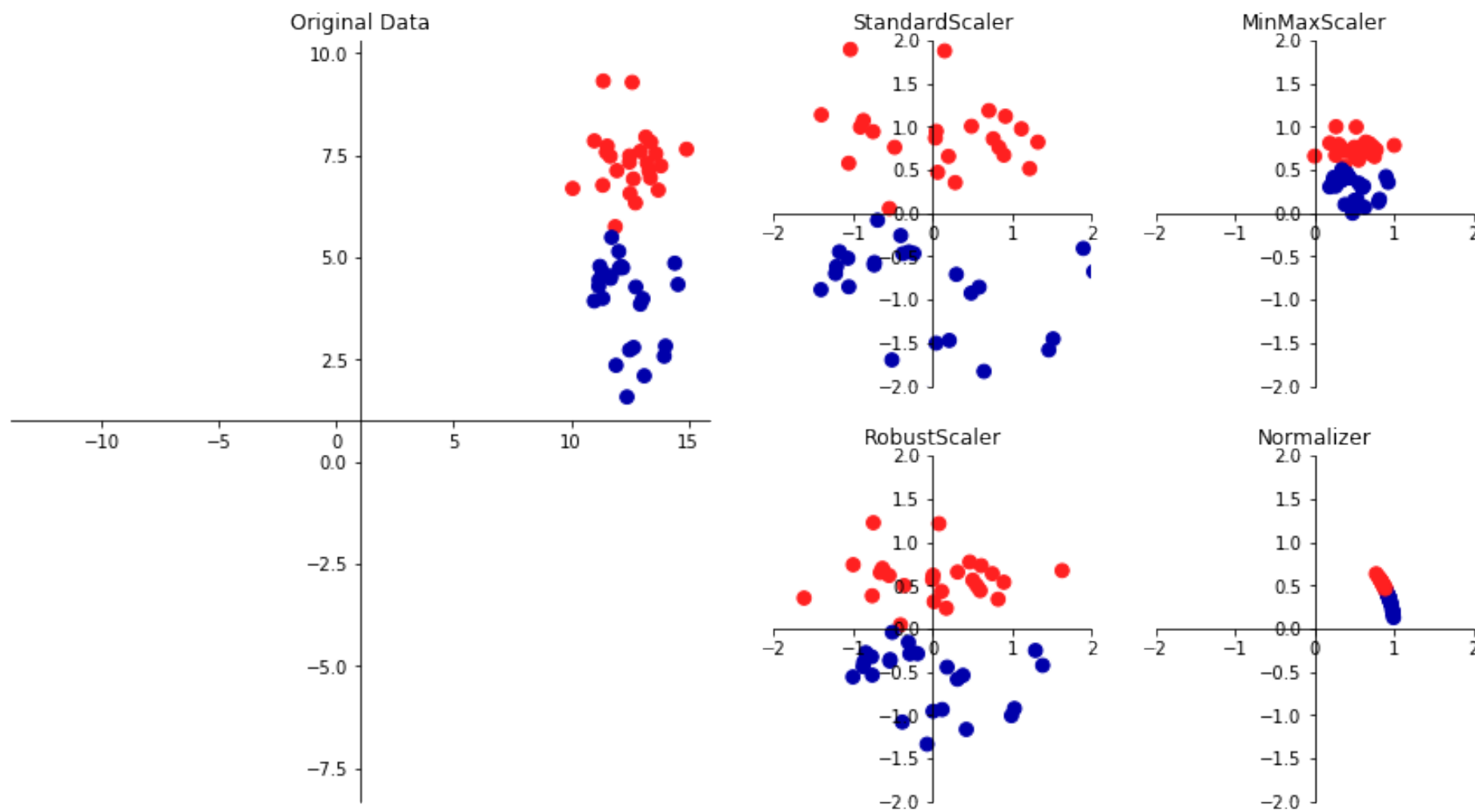
ВВЕДЕНИЕ В МАШИННОЕ ОБУЧЕНИЕ

О КРОСС-ВАЛИДАЦИИ, КОНТЕЙНЕРАХ И ПЕРЕМАСШТАБИРОВАНИИ
ДАННЫХ

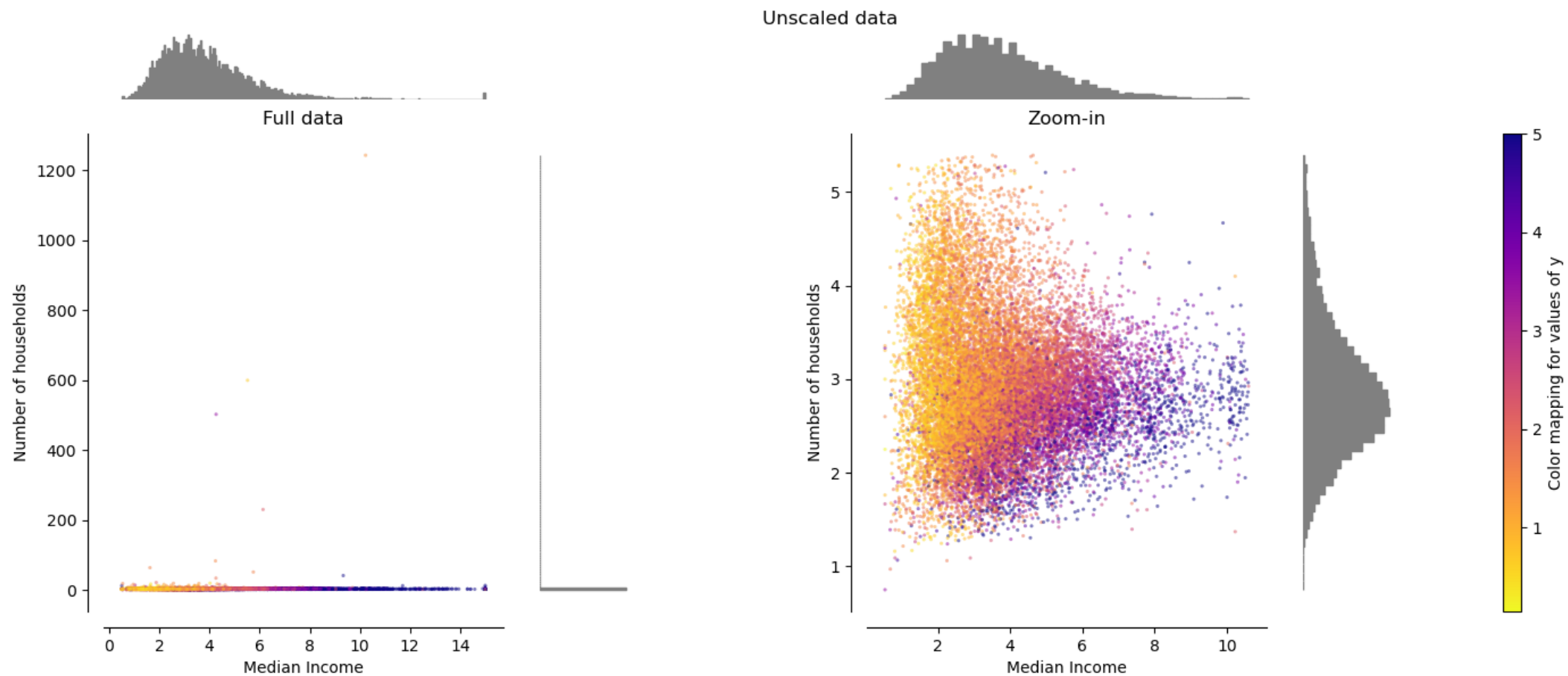
Москва, 2026

Материалы занятия <https://github.com/kshilin/machine-learning>

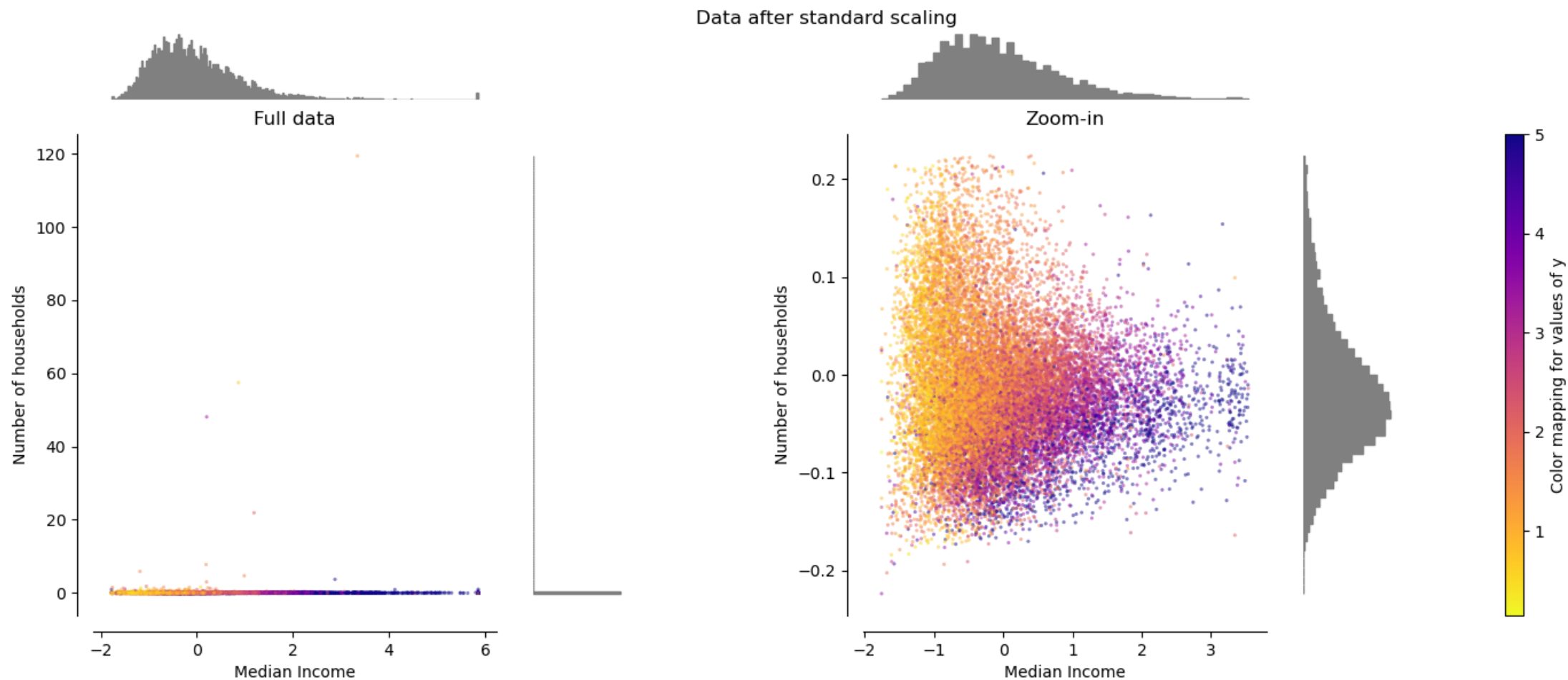
РАНХиГС



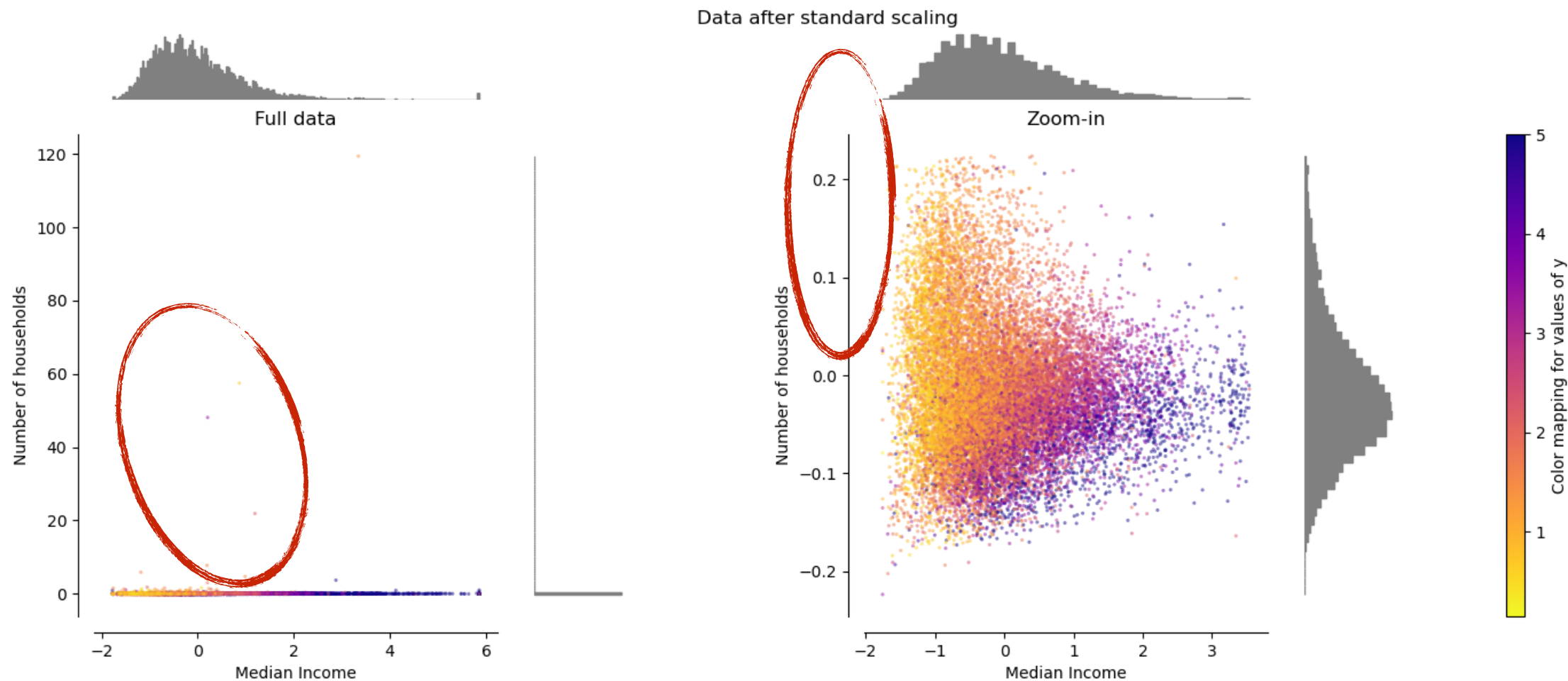
Исходные данные



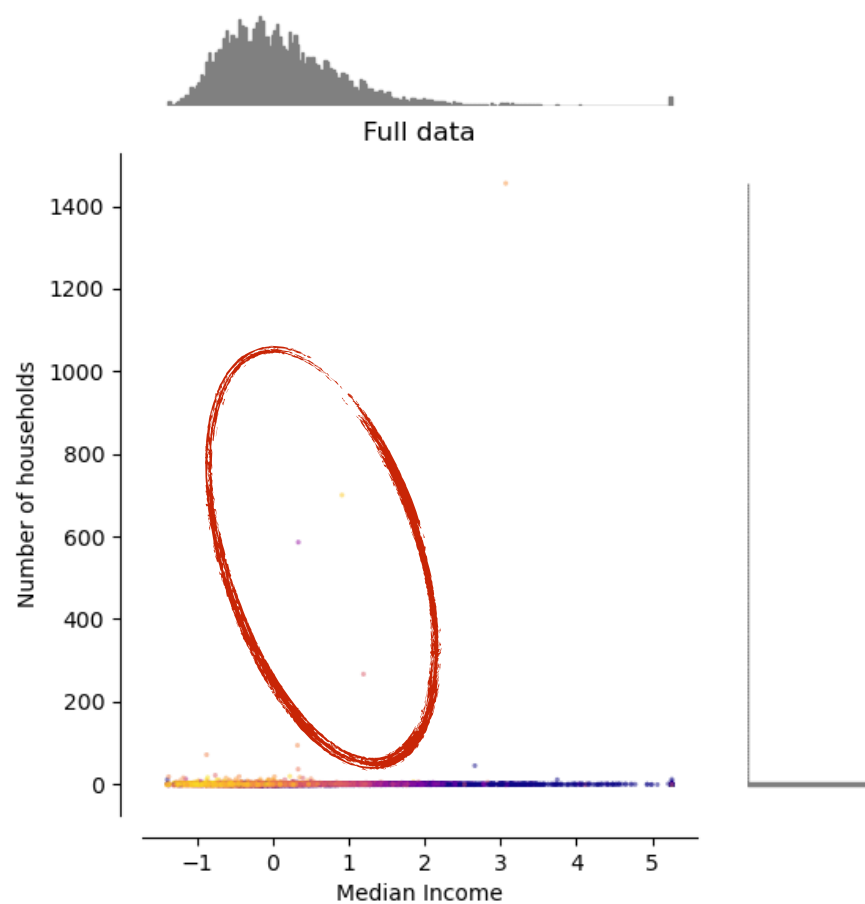
StandardScaler



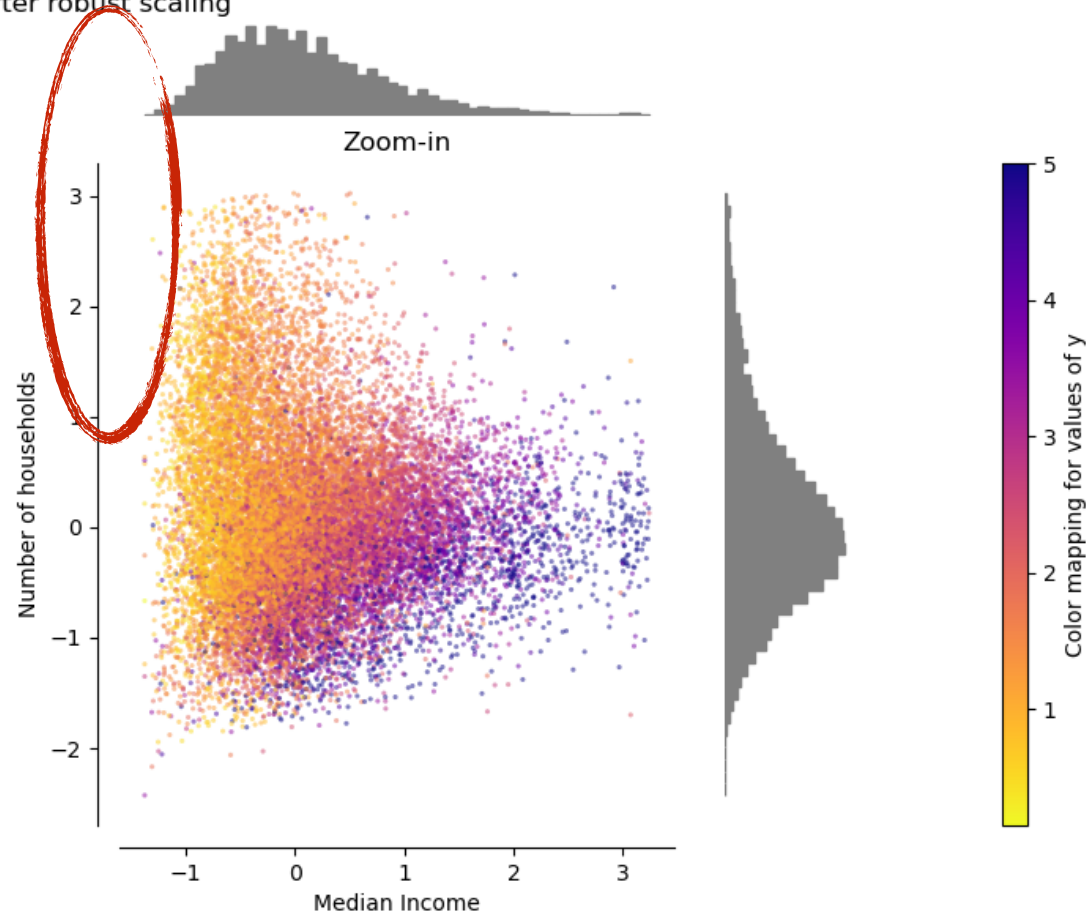
StandardScaler и выбросы



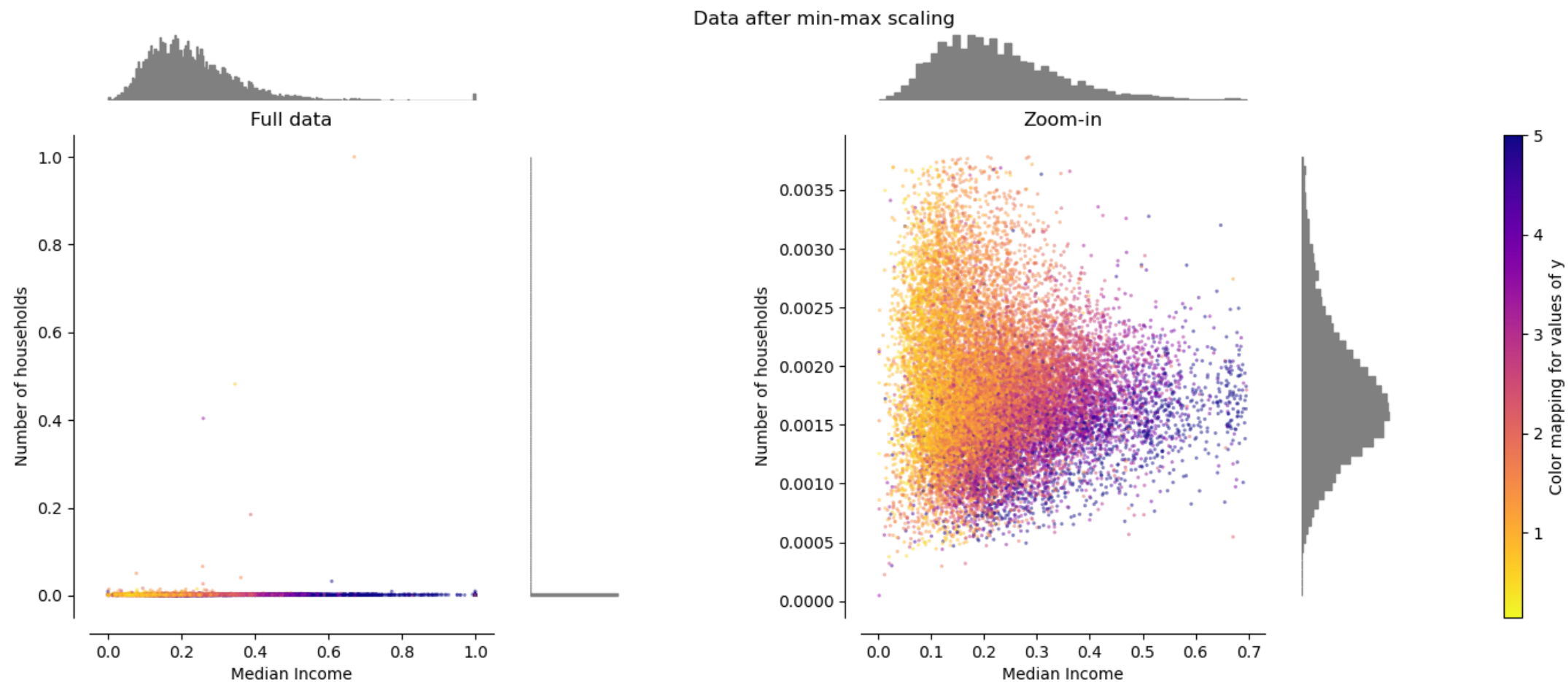
RobustScaler



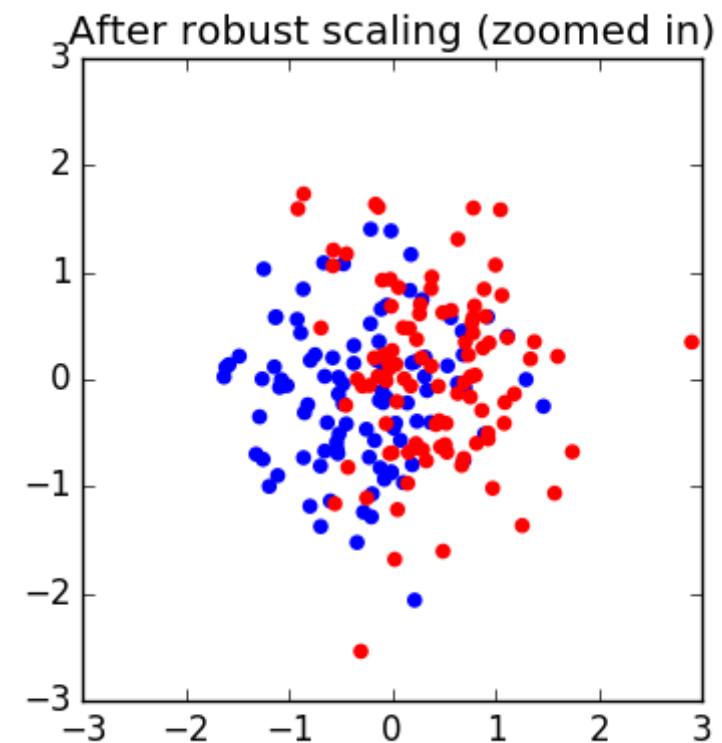
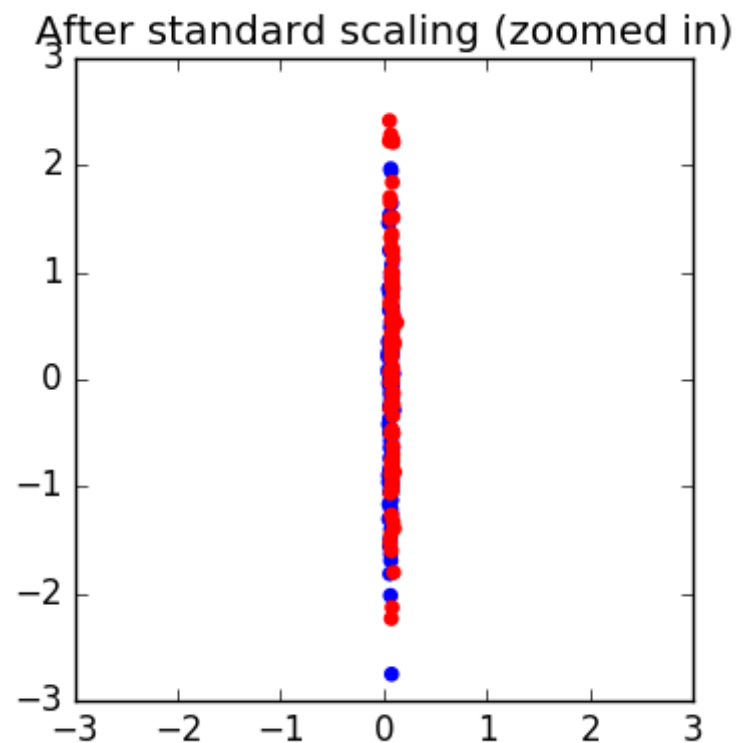
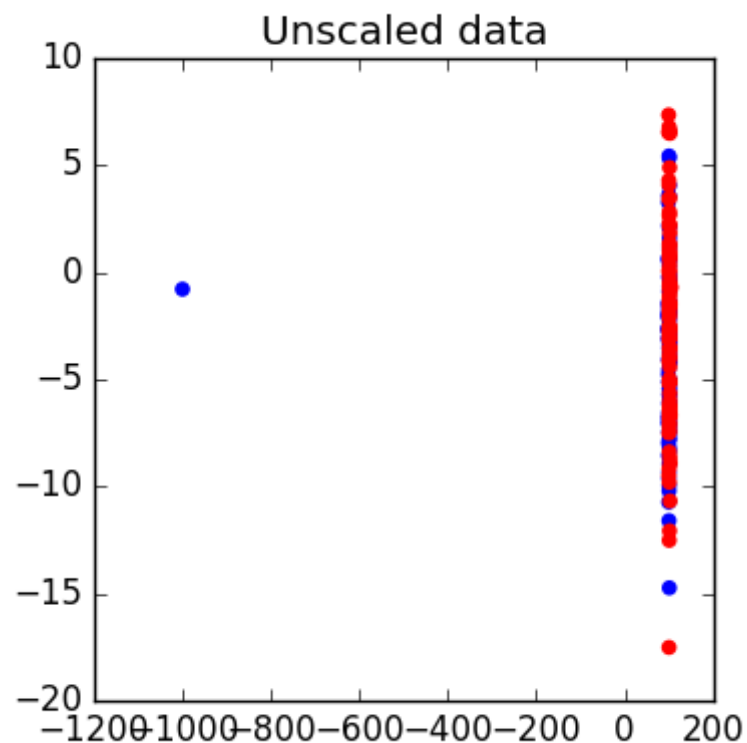
Data after robust scaling



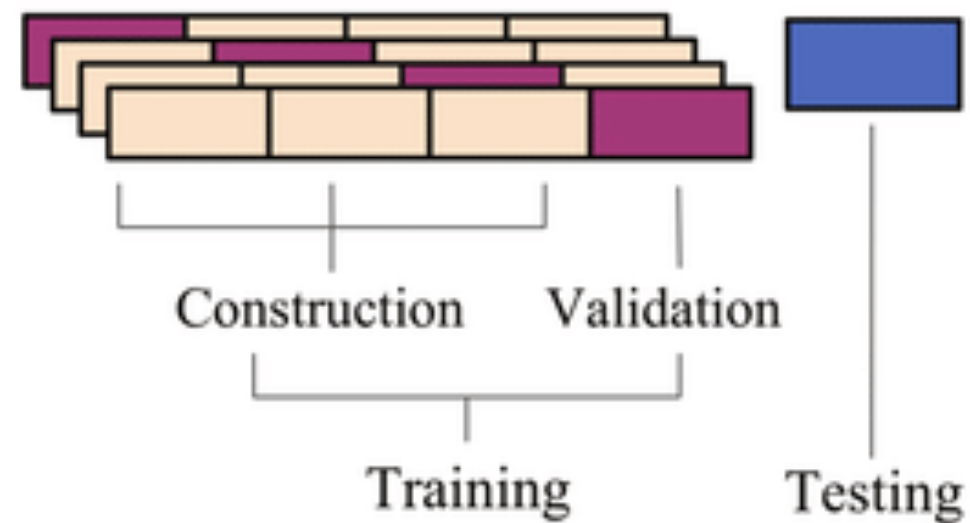
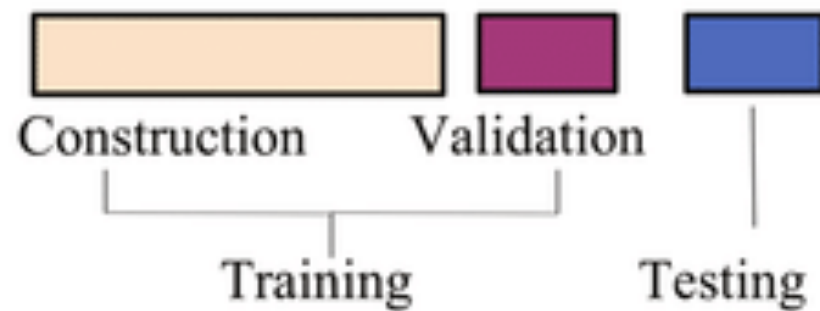
MinMaxScaler



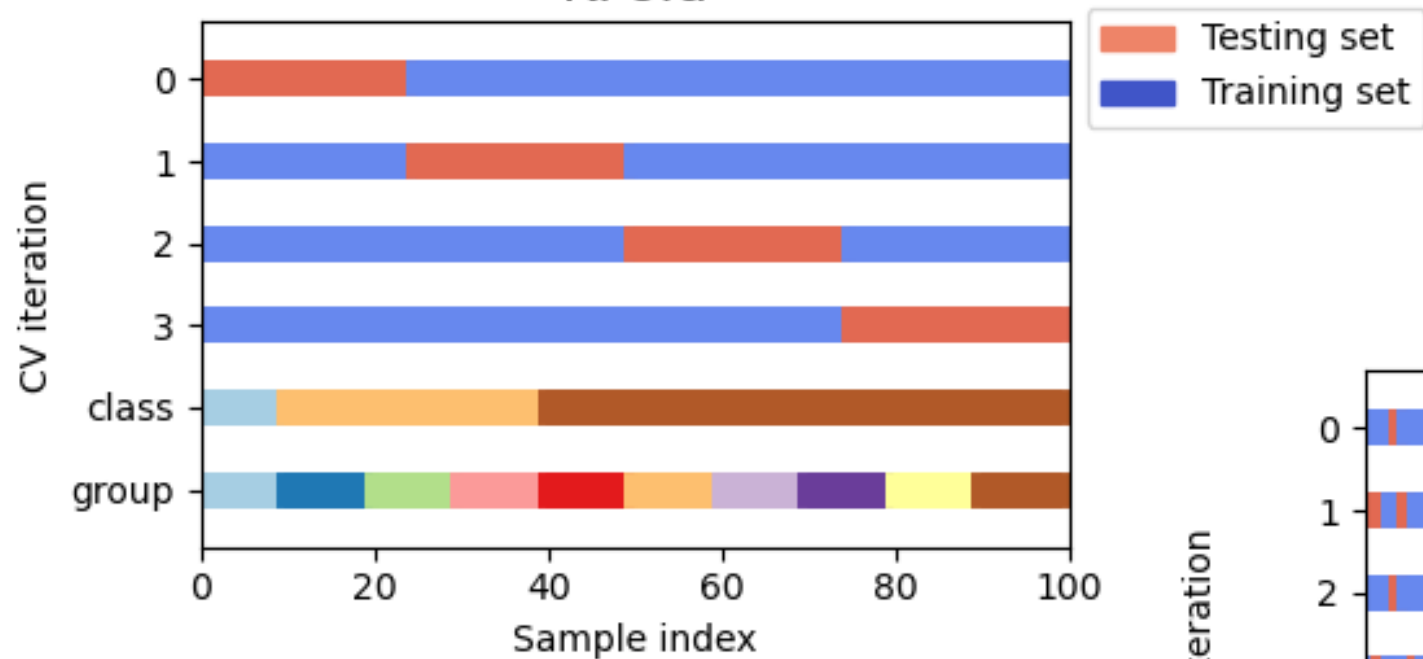
Влияние выбросов



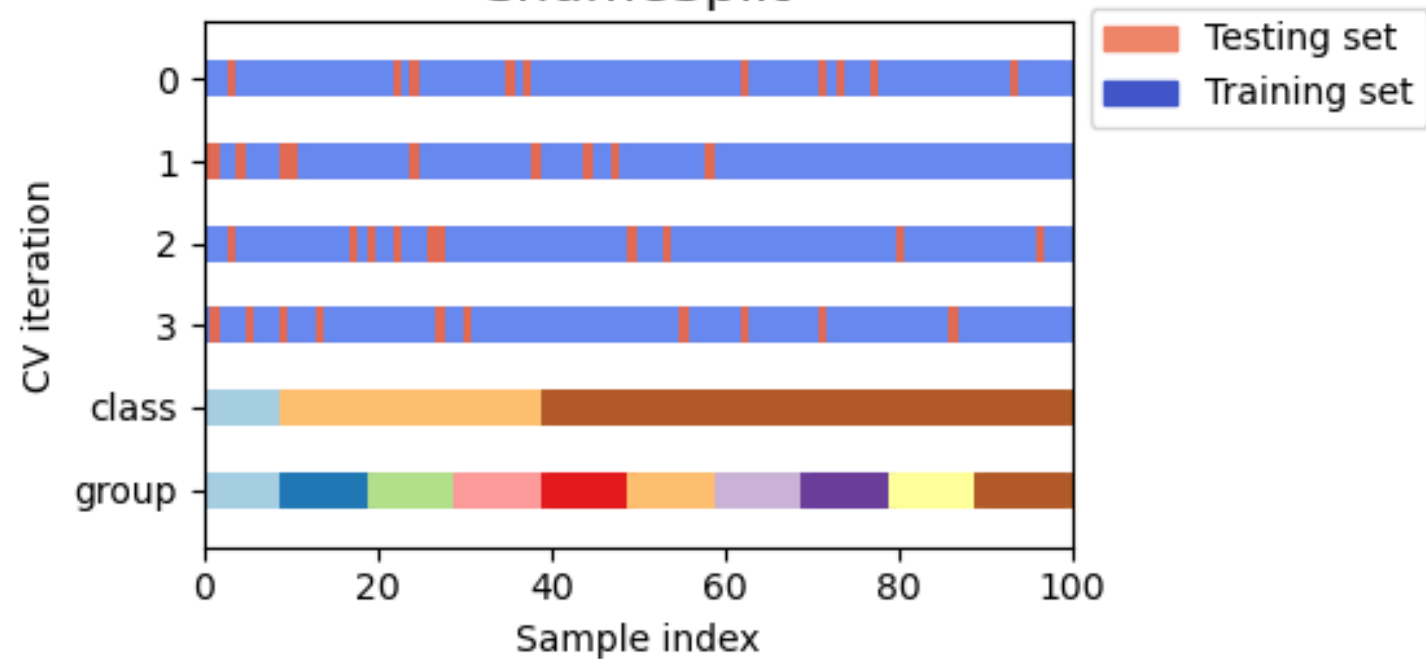
Кросс-валидация



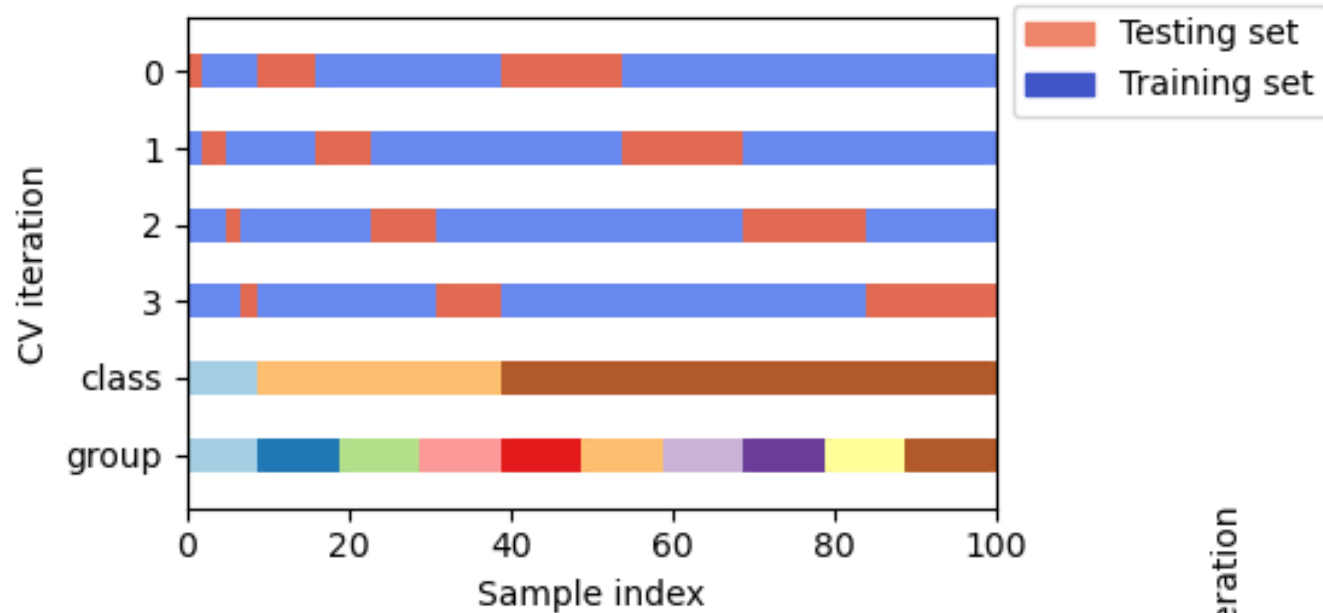
KFold



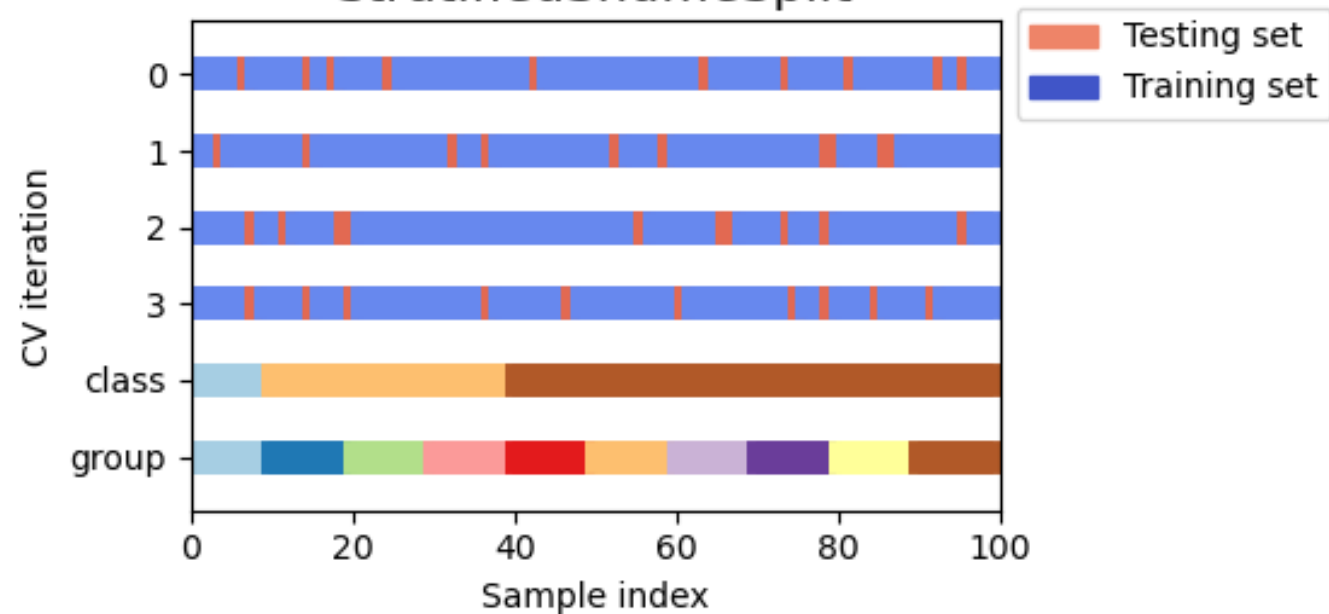
ShuffleSplit



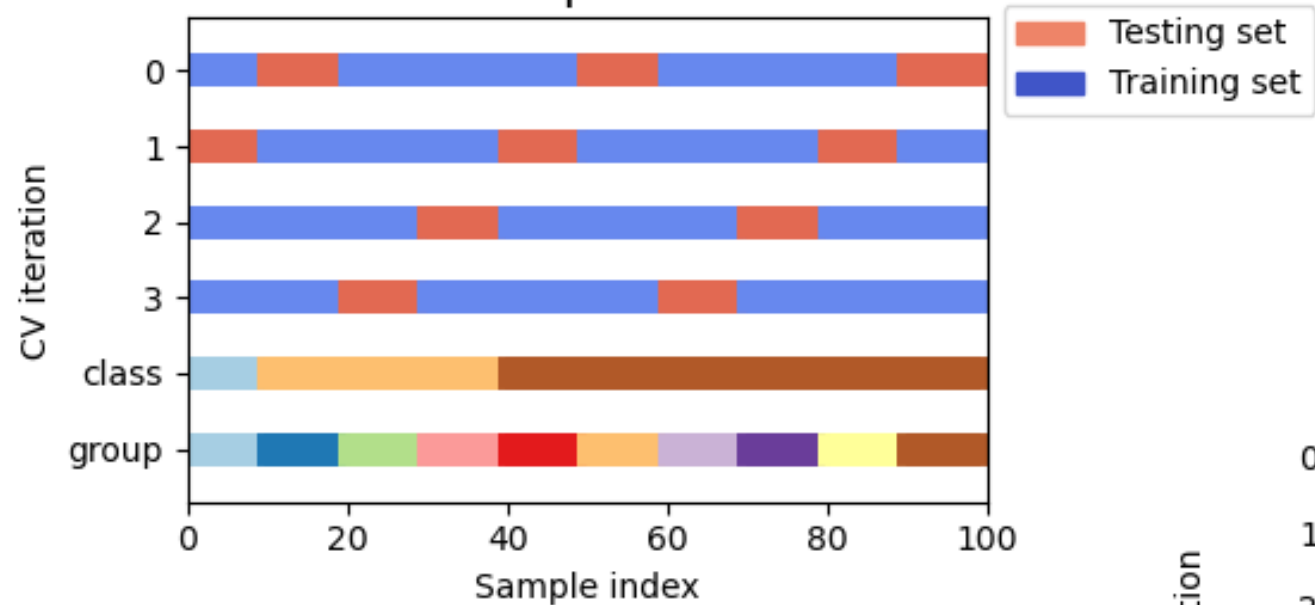
StratifiedKFold



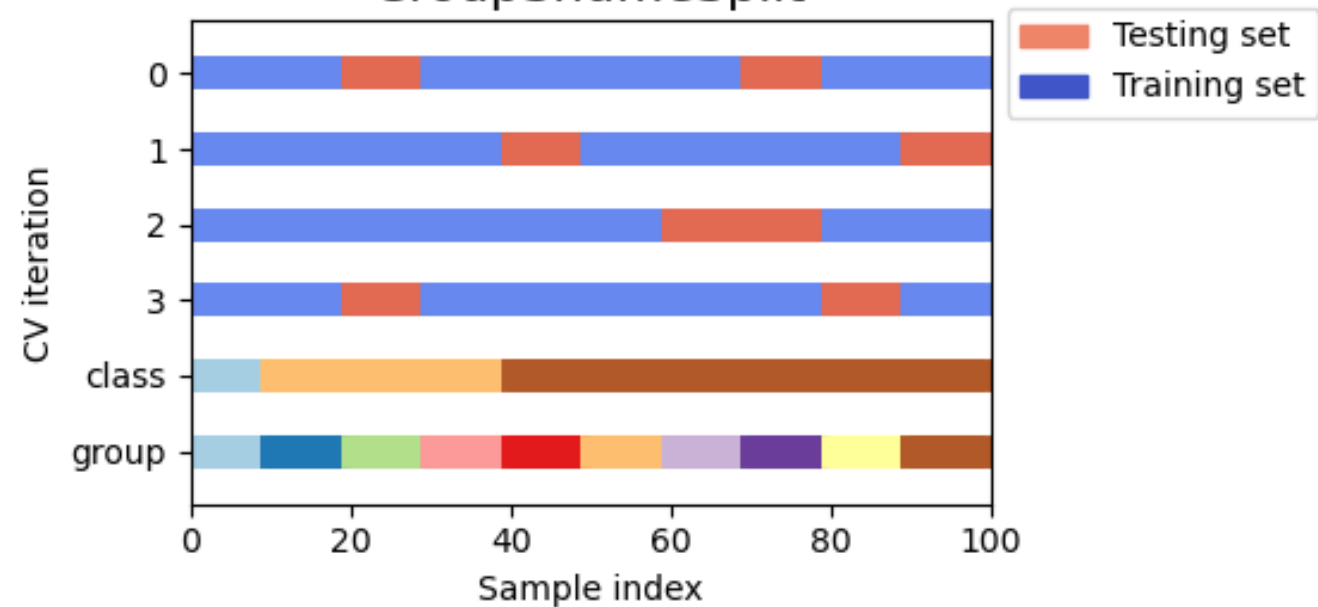
StratifiedShuffleSplit

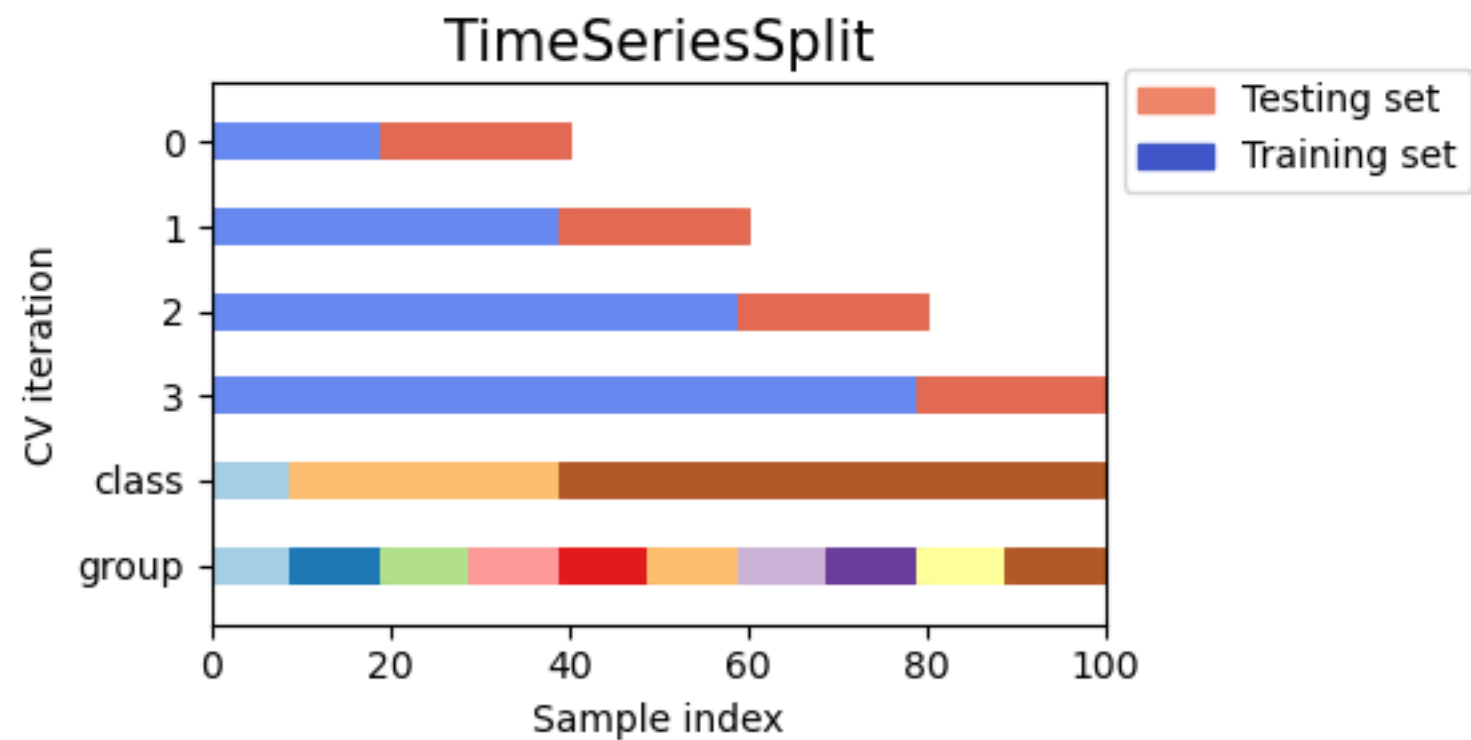


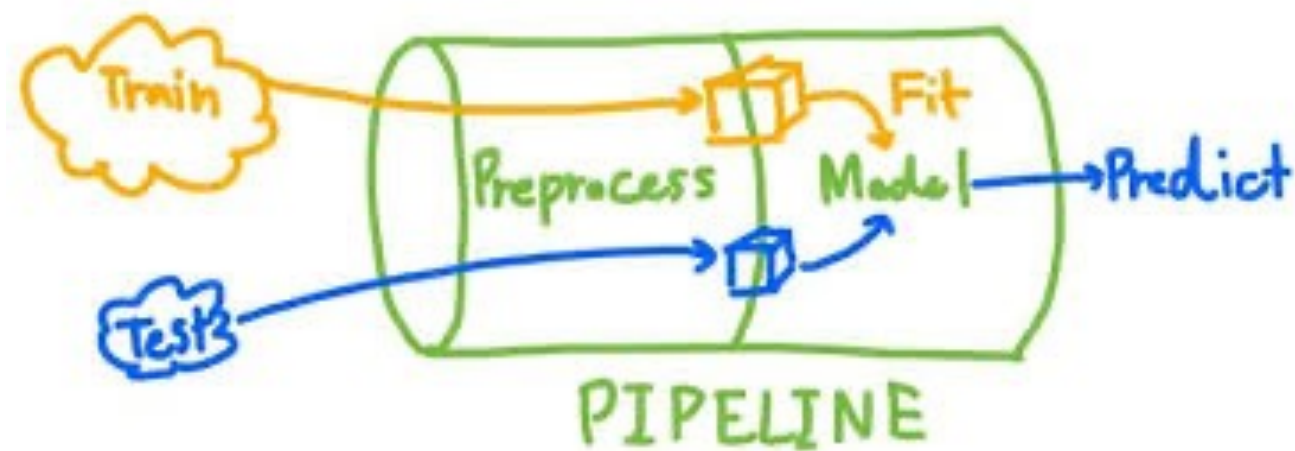
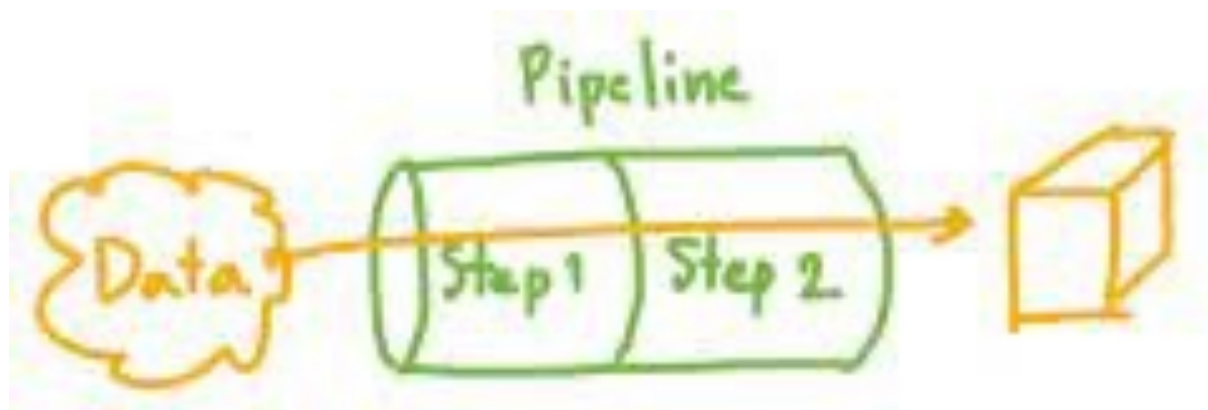
GroupKFold



GroupShuffleSplit









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СПАСИБО ЗА ВНИМАНИЕ!

Москва

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